

Cookie Monster's Growing Cookie Jar

Where the Number e Comes From

November 4, 2026 · Nathaniel

Name: _____ Date: _____

The story

Cookie Monster has a **magic cookie jar**. It starts with **256 cookies**. Once a day, the jar **grows**: it makes as many new cookies as it already has. But only if Cookie Monster does not eat any! *"Me wait. Me wait. Me VERY hungry, but me wait!"*

Then **Grover** has an idea: *"What if the jar grows a LITTLE bit, MANY times a day? Would that make MORE cookies?"* Let's find out. We will discover a famous number called e.

Growing once

Once a day, the jar adds as many cookies as it has. That means it **doubles**.

Question 1. The jar has 256 cookies. It adds 256 more. $256 + 256 =$ _____



start: 256 end: _____

Grover's idea: grow two times, a half each time

*"Grow **two** times a day, but each time add only **half** of what is in the jar!"* Remember: a half is one of two equal parts.

Question 2. Fill in the table. The first row is done for you.

Grow	Cookies in jar	Add half	New total
1st time	256	128	384
2nd time	384		

Question 3. Two halves made _____ cookies. Is that **more** than growing once (Question 1)?

_____

Grow four times, a quarter each time

*"Even MORE times! Grow **four** times, and each time add a **quarter**!"*

> **Quarter trick:** a quarter is **half of a half**. To find a quarter of 256, take half (128), then half again (64). So a quarter of 256 is 64.

Question 4. Use the quarter trick to fill in the table.

Grow	Cookies in jar	Add a quarter	New total
1st time	256	64	320
2nd time	320		
3rd time			
4th time			

Question 5. Fill in what The Count found so far. *"Ah ah ah! Let me count the cookies!"*

How the jar grows	Cookies at the end
1 time, add all of it	
2 times, add a half	
4 times, add a quarter	

Question 6. More, smaller steps made _____ cookies. (more or fewer?)

What if we grow MANY times?

The steps get too tiny to do by hand, so The Count used a computer. Here is what he found:

How many tiny steps in a day	Cookies at the end
8	about 657
100	about 692
1,000	about 696
1,000,000 (one million!)	about 696

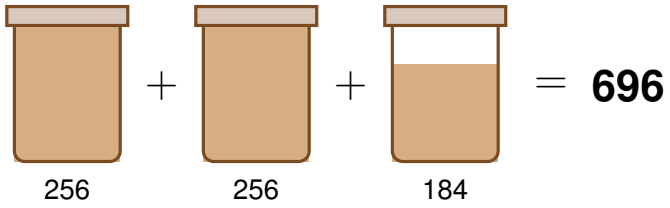
Question 7. Does the jar keep growing bigger and bigger forever, or does it stop near one number?

_____

> More steps make more cookies, but not forever! The jar gets closer and closer to **about 696 cookies**, and never goes past it. Grover's idea helps, but only up to a point.

How many jars full?

The jar started with 256 cookies. Call that **one jarful**. How many jarfuls is 696 cookies?



Question 8. Two full jars is $256 + 256 =$ _____ cookies. The last jar is a little less than _____ full. (half or three-quarters?)

That is **2 jarfuls and a bit more than 7 tenths of another**: about $\mathbf{2.72}$ jarfuls. Mathematicians call this number $\boldsymbol{e}$:

$$e \approx 2.72$$

It is the number you get when something **grows in tiny, tiny steps**. The jar ends with e jarfuls, no matter how many cookies it starts with!

Question 9. A different magic jar starts with 100 cookies. About how many cookies will it have after growing in tiny steps? (Hint: e jarfuls of 100.) about _____

The Count's Math Club: write it like a scientist

The number e is named after **Leonhard Euler** (say "OY-ler"), who studied it. It goes on forever without repeating: $e = 2.71828\ldots$ Scientists write "grow by all of it, in n tiny steps" like this:

$$\left(1 + \frac{1}{n}\right)^n \longrightarrow e \quad \text{when } n \text{ gets very big.}$$

The little n up high means "do it n times." With $n = 4$ you did $\left(1 + \frac{1}{4}\right)^4$: add a quarter, four times!

Coming up: The Cookie Circle

Here is Euler's big surprise. If the jar grows in tiny steps, you get e. But if each tiny step is a tiny **turn**, using j from The Count's Square Cookies, you do not grow at all. You walk around a **circle**! That is why e and j meet in Euler's famous equation, $e^{j\pi} + 1 = 0$.

For teachers: this is compound growth at a 100% rate: $256\,(1{+}1/n)^n$ gives 512, 576 and 625 for $n = 1, 2, 4$, and tends to $256\,e \approx 695.9$ (the table values are rounded). The 256-cookie start keeps every hand-computed step a whole number. The "Coming up" box previews $e^{j\theta} = \lim_{n\to\infty}(1 + j\theta/n)^n$: each factor $1 + j\theta/n$ is, to first order, a rotation by θ/n, so the product rotates by θ. Sesame Street characters appear by name only, for classroom use; this worksheet is not affiliated with or endorsed by Sesame Workshop.

Answer key

Question	Answer
1	512
2	2nd time: 384, add 192, total 576
3	576; yes, more than 512
4	2nd: 320, add 80, total 400 3rd: 400, add 100, total 500 4th: 500, add 125, total 625
5	512; 576; 625
6	More
7	It stops near one number (about 696)
8	512; three-quarters (184 out of 256 is about 0.72)
9	About 272 (2.72 jarfuls of 100)